

Aayush Swami

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SUMMARY

CS junior with experience building scalable backend systems, fast serverless APIs, and AI tools using Python, TypeScript, and cloud infrastructure. Proven track record developing scalable ML pipelines, agent-based systems, and full-stack web tools. Experienced working with cross-functional teams in both research and startup environments. Actively seeking an SWE & AI/ML internship for Summer–Fall 2026 and open to relocation.

EDUCATION

Bachelor of Science in Computer Science
Arizona State University, Tempe, AZ

Graduating May 2027

GPA: 3.88

Relevant Coursework: Data Structures & Algorithms, Operating Systems, Distributed Software Development, Principles of Programming Languages, Software Engineering, Computer Organization, Object-Oriented Programming, Human–Computer Interaction, Applied Linear Algebra, Probability & Statistics, Discrete Math

TECHNICAL SKILLS

Languages:	Python, TypeScript, Java, C/C++, Go, JavaScript, SQL, Swift, MATLAB, HTML/CSS
Frameworks:	FastAPI, MERN, PyTorch, TensorFlow, Scikit-learn, Hugging Face, NumPy, Pandas, OpenCV, Matplotlib
Databases:	PostgreSQL, MongoDB, MySQL, Firebase, InfluxDB
Tools:	Cloudflare Workers, Cloudflare Pages, Docker, Kubernetes, Git, Edge Runtime, DNS APIs
Cloud/DevOps:	AWS, Google Cloud, Render, GPU Clusters, CI/CD Pipelines
Methodologies:	Agile Scrum, Kanban, RESTful API Development, Version Control, Data Visualization

PROFESSIONAL EXPERIENCE

ML Researcher – Rolston Lab, Arizona State University

Aug 2024 – Present

- Built a BLIP-2 + PyTorch pipeline to extract microscopy/sensor data, enabling automated preprocessing for model training.
- Optimized GPU usage on ASU's supercomputing clusters and redesigned inference flow, cutting model latency by 65%.
- Achieved 0.94 R^2 with GPR model for battery forecasting, enhancing prediction accuracy and downstream reliability.
- Wrote full technical documentation to support reproducible training, validation, and deployment of AI models.

Software Engineer Intern – Spectra Education Consultancy

May 2025 – Aug 2025

- Developed an AI support chatbot using Python, NLP, LLMs, and Agentic-AI style workflows, cutting response time by 60%.
- Re-architected MERN + TypeScript backend and introduced Docker-based CI/CD pipeline, improving platform uptime to 99%.
- Added ML recommendation system that personalized content and boosted student engagement by 40%.
- Wrote clear documentation for our APIs and workflows to make updates easier for the engineering and content teams.

Software Developer – Odyssey Rover Project

Jan 2025 – May 2025

- Engineered real-time telemetry dashboard (TypeScript, Python) to stream and visualize rover sensor data with low-latency updates.
- Integrated LiDAR, ultrasonic, and environmental sensors with Raspberry Pi to support autonomous rover navigation and control.
- Developed LLM-based diagnostic system to detect and respond to hardware anomalies, enabling semi-autonomous system recovery.

PROJECTS

AI-Powered Trading Agent, Financial Market Prediction Platform – Developer

Jan 2025 – Present

- Architected Python + FastAPI + PostgreSQL system for real-time ML trading predictions with GPU acceleration and serverless APIs.
- Integrated BLIP-2 for multimodal chart analysis and combined it with technical indicators like RSI, EMA, and Supertrend.
- Experimented with multiple ML algorithms and feature-engineering techniques, reaching \$1.31M simulated profit from \$100.
- Designed modular agent workflows (inspired by LangChain/LangGraph) to speed up iteration, testing, and pipeline improvements.

Auto Cover-Letter Generator & Greenhouse Auto-Apply Tool – Developer

Oct 2025 – Present

- Integrated a **Python web-scraper** (Requests, BeautifulSoup) to extract skills and role details from any job posting.
- Used LLMs, NLP techniques, and an ATS keyword-matching model to generate personalized, resume-aligned cover letters.
- Built serverless AI microservices with Cloudflare Workers AI, Durable Objects, GO and KV for ultra-low-latency inference.
- Created an automated Greenhouse apply workflow using Selenium/Playwright to streamline repetitive application steps.

Weather & Café Finder iOS App, iOS App (Swift) – Developer

Oct 2024 – Dec 2024

- Built and deployed an iOS app in Swift that suggests nearby cafés and clothing choices using real-time weather and geolocation APIs.
- Integrated Firebase backend for user authentication and preference storage, improving recommendation accuracy and app reliability.

ACTIVITIES

Core Team Member, Google Developer Group, Arizona State University

Aug 2024 – Present

- Organized and led technical workshops on ML and web development, helping over 100 students build real-world projects.

Adobe Student Ambassador, Adobe

Jan 2025 – Present

- Participated in closed testing of new AI features in Adobe Express app, providing feedback on usability, responsiveness, and UX design.